

IS105 - ENTERPRISE ARCHITECTURE

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SEQUENCE DIAGRAM



AT THE END OF THIS LESSON, YOU SHOULD BE ABLE TO:

- Familiarize with the UML Sequence diagram.
- Be conscious how important UML Sequence Diagram is for visualizing and documenting software systems.

SEQUENCE DIAGRAM

- A sequence diagram is a type of interaction diagram within the UML that visually represents the interactions between objects over time.
- It illustrates the sequence of messages exchanged between objects and the order in which methods or operations are called.
- Sequence diagrams are sometimes known as event diagrams or event scenarios.

BENEFITS OF SEQUENCE DIAGRAMS



BENEFITS

- Represent the details of a UML use case.
- Model the logic of a sophisticated procedure, function, or operation.
- See how objects and components interact with each other to complete a process.
- Plan and understand the detailed

BASIC SYMBOLS AND COMPONENTS



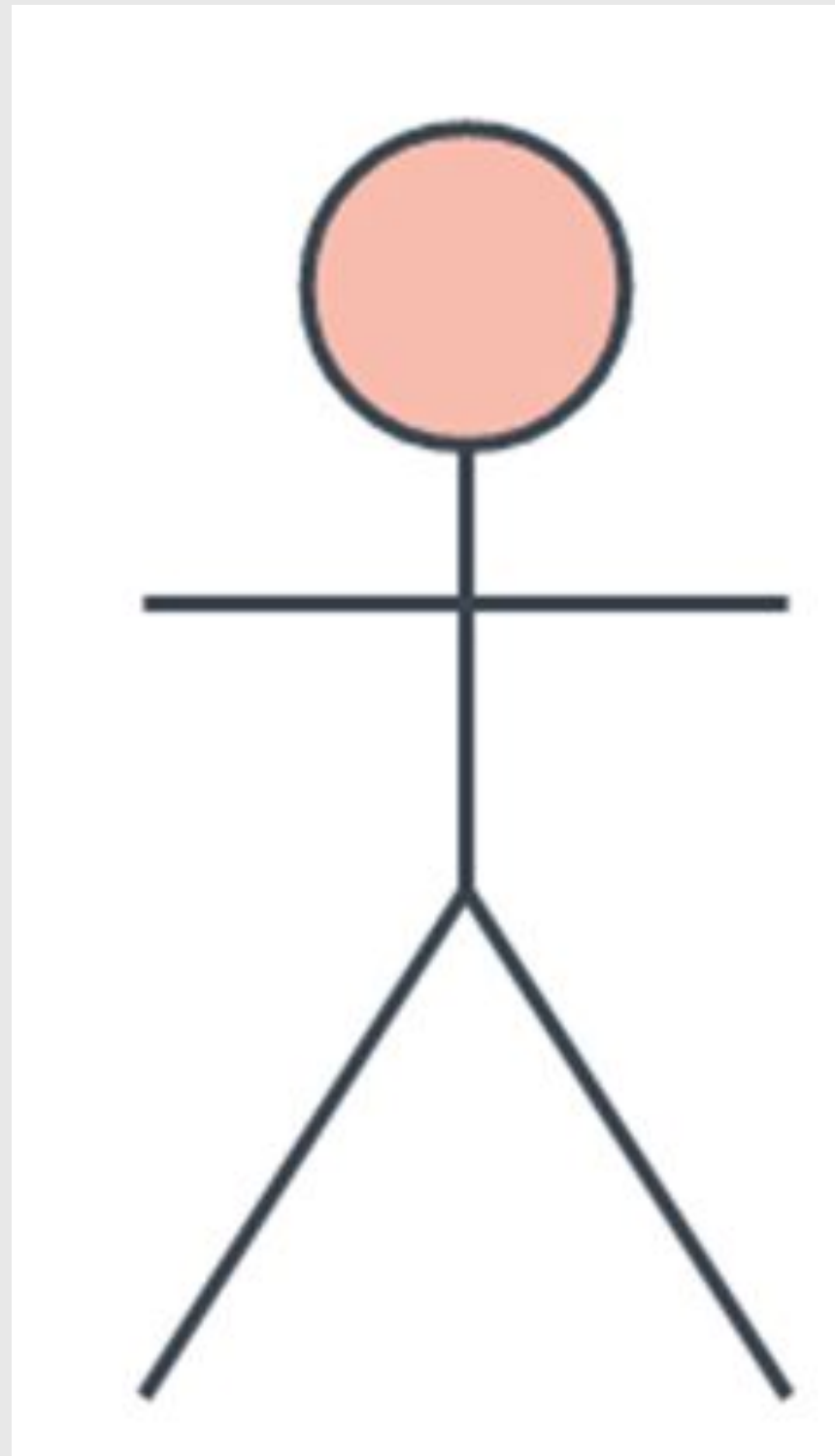
OBJECT

- Represents a class or object in UML. The object symbol demonstrates how an object will behave in the context of the system. Class attributes should not be listed in this shape.



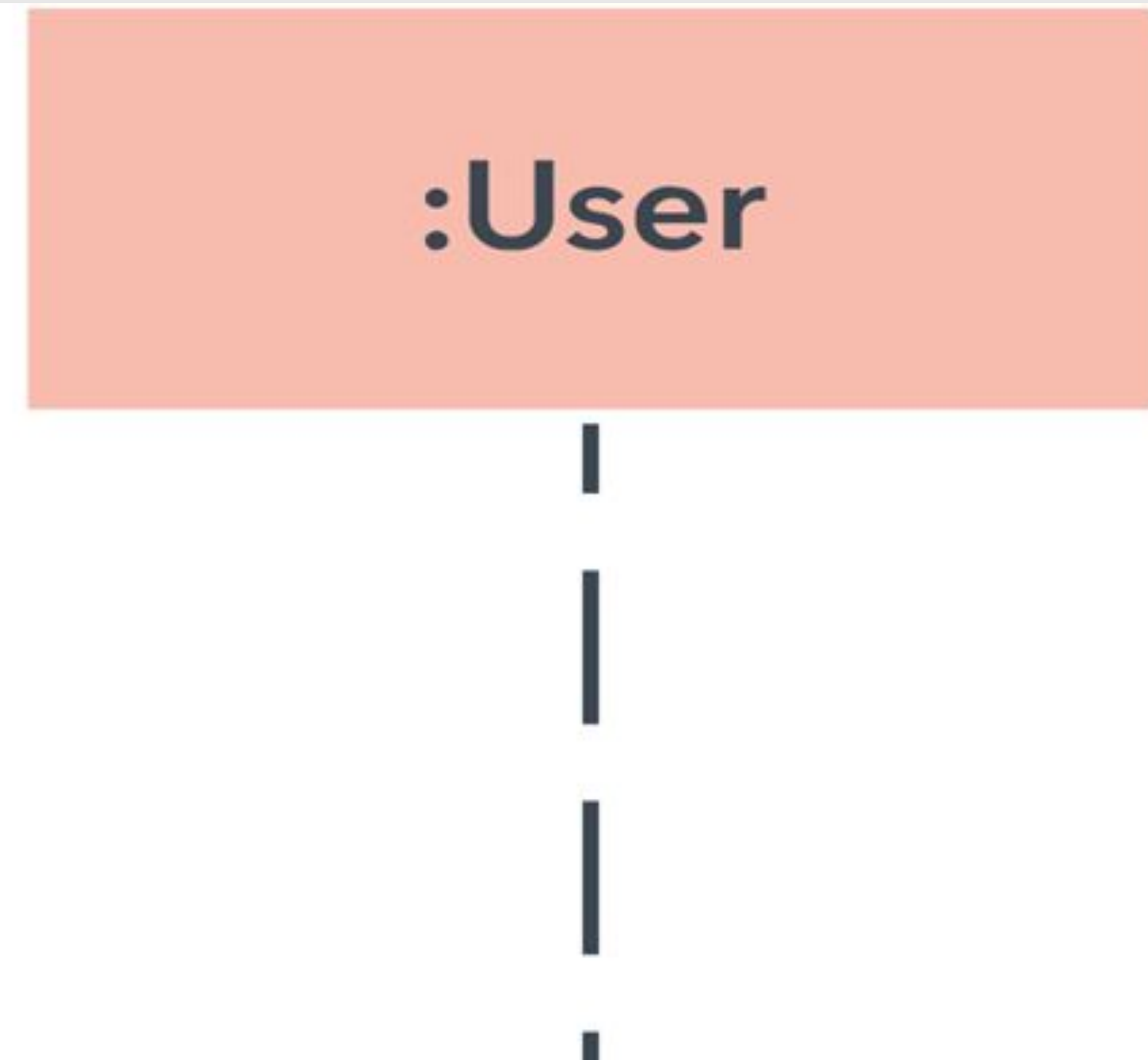
ACTOR

- Shows entities that interact with or are external to the system.



LIFELINE

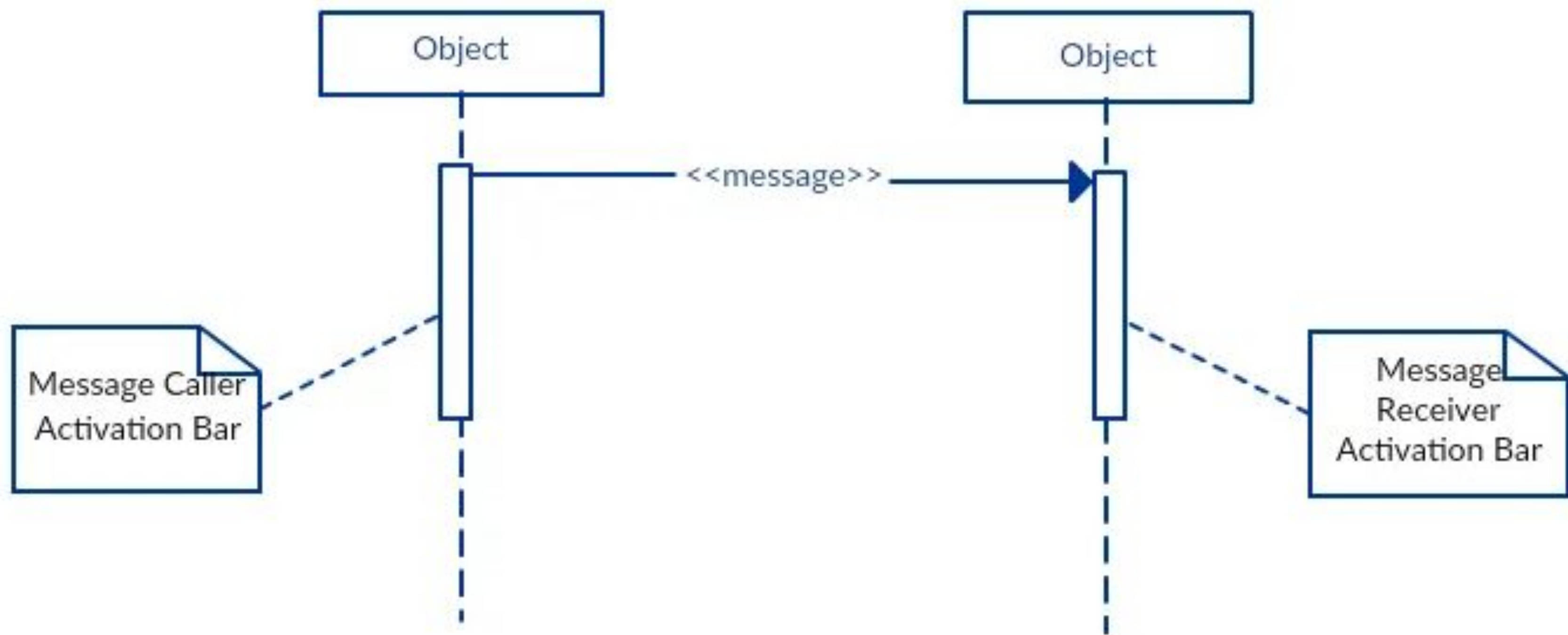
- Represents the passage of time as it extends downward. This dashed vertical line shows the sequential events that occur to an object during the charted process.
- Lifelines may begin with a labeled rectangle shape or an actor symbol.



ACTIVATION BOX

- Represents the time needed for an object to complete a task.
- The longer the task will take, the longer the activation box becomes.





COMMON MESSAGE SYMBOLS



SYNCHRONOUS

MESSAGE Represented by a solid line with a solid arrowhead. This symbol is used when a sender must wait for a response to a message before it continues. The diagram should show both the call and the reply.



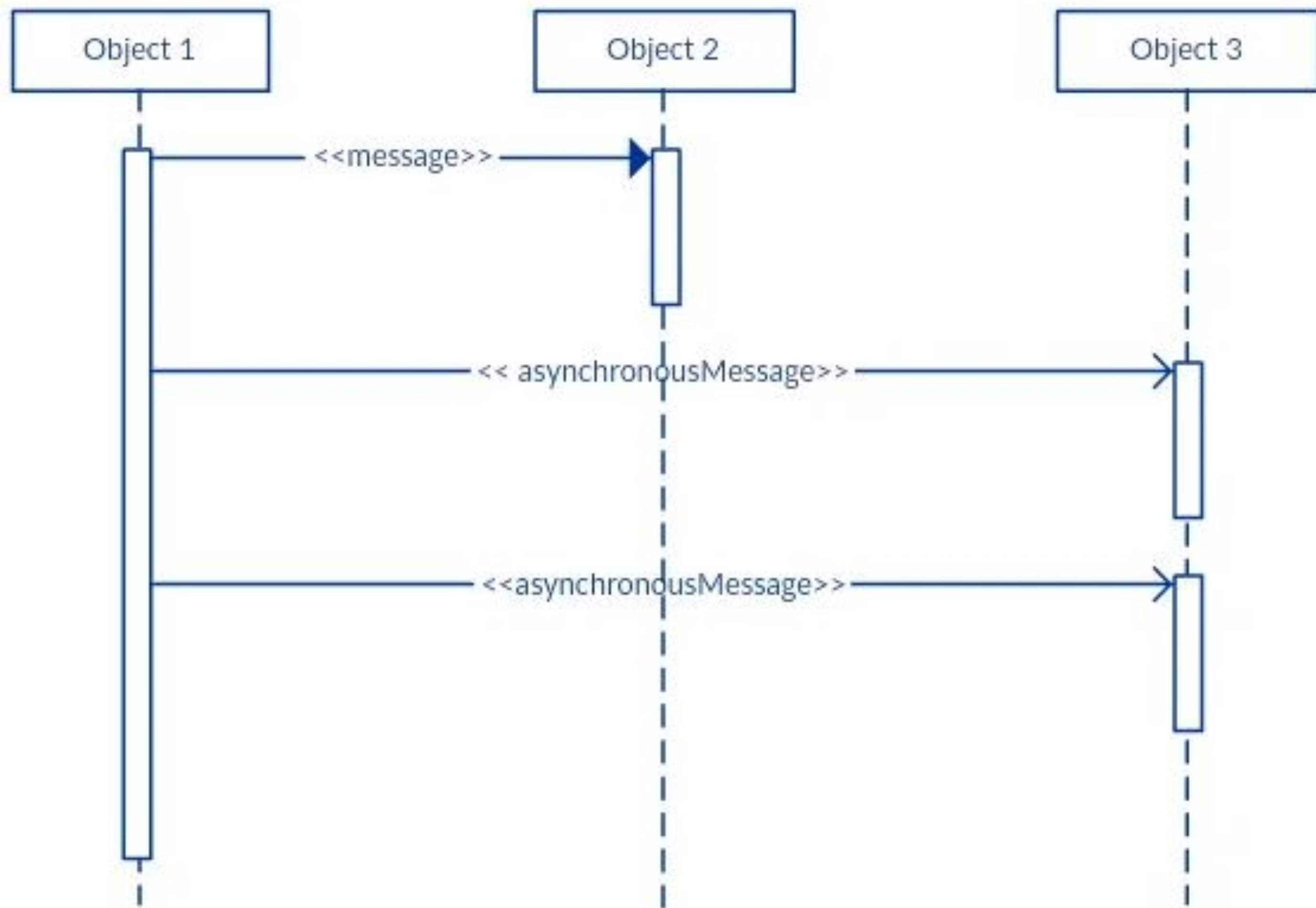
ASYNCHRONOUS

MESSAGE

Represented by a solid line with a lined arrowhead.

Asynchronous messages don't require a response before the sender continues. Only the call should be included in the diagram.





ASYNCHRONOUS RETURN

MESSAGE

- Represented by a dashed line with a lined arrowhead. The sender can continue with other tasks while the receiver processes the message.

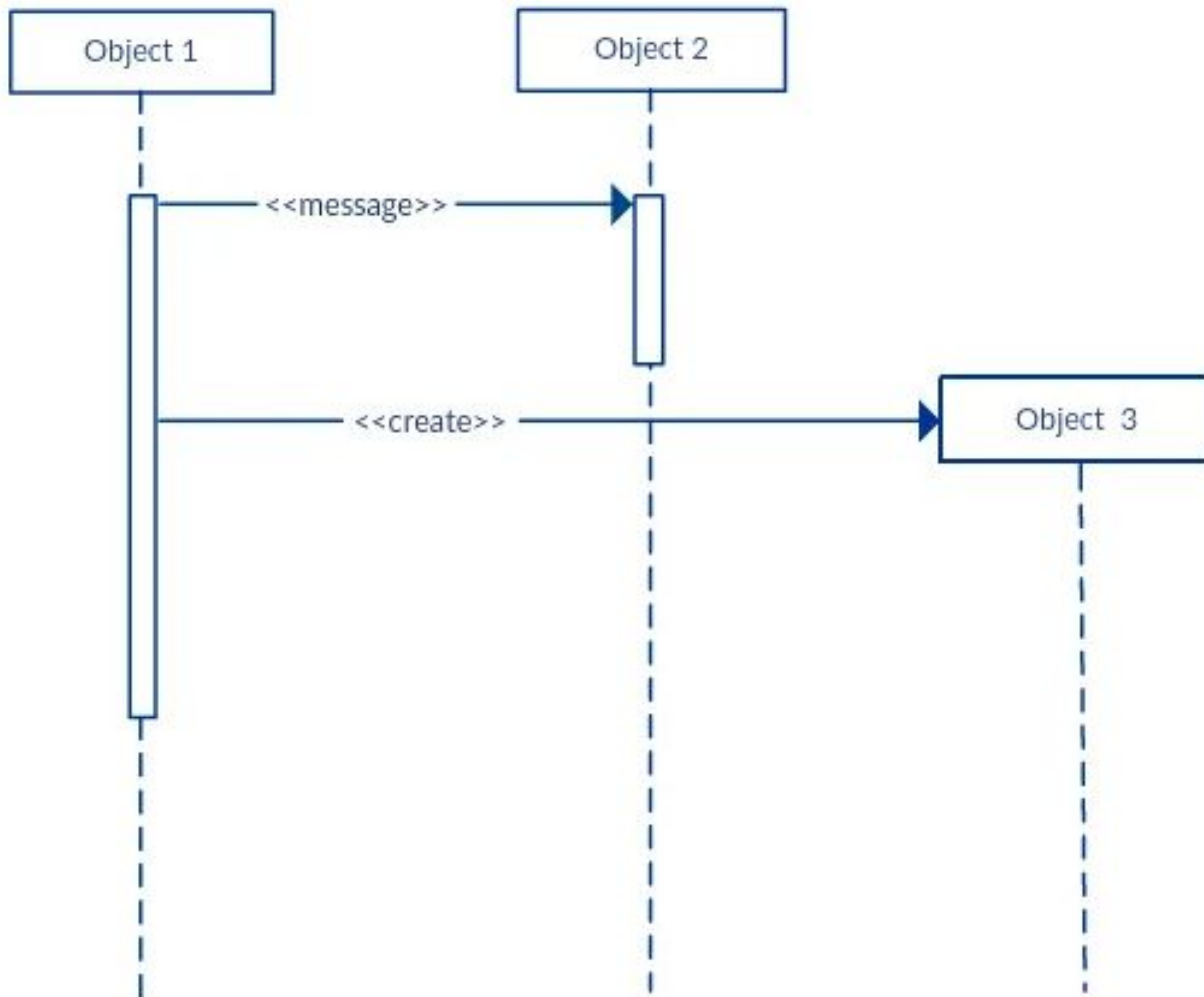


ASYNCHRONOUS CREATE

MESSAGE

- Represented by a dashed line with a lined arrowhead. This message creates a new object.





REPLY MESSAGE

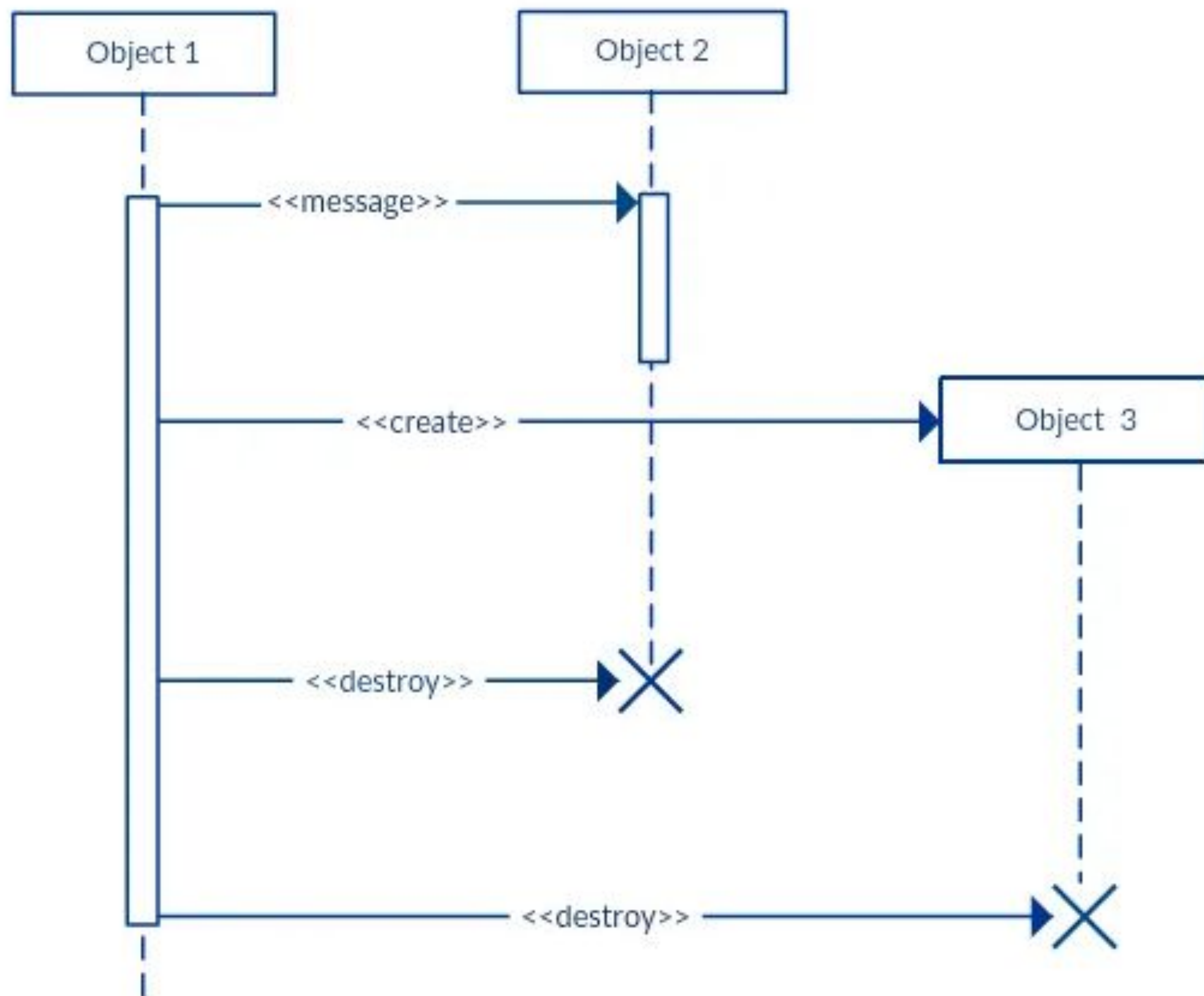
- Represented by a dashed line with a lined arrowhead, these messages are replies to calls.



DELETE MESSAGE

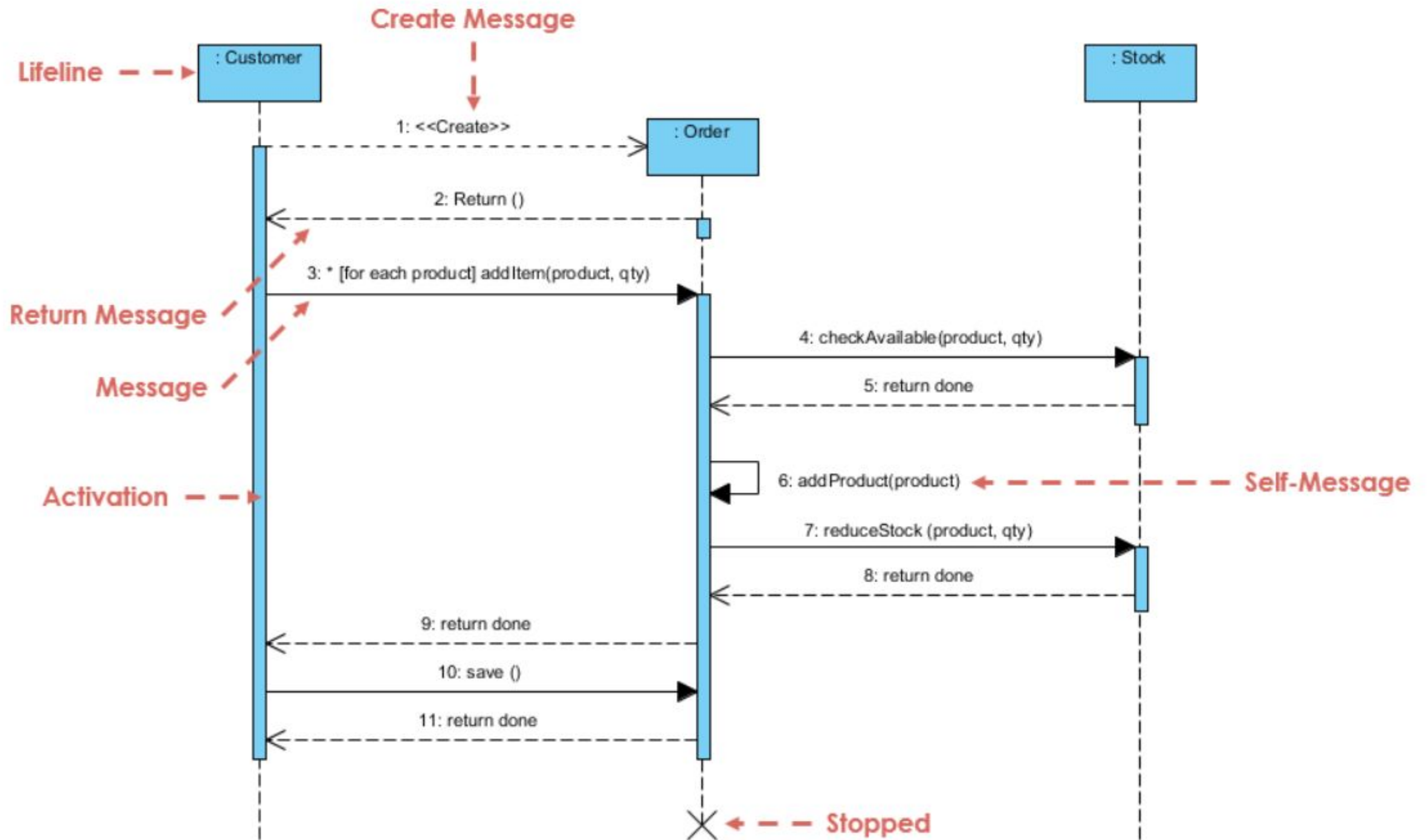
- Represented by a solid line with a solid arrowhead, followed by an X. This message destroys an object.

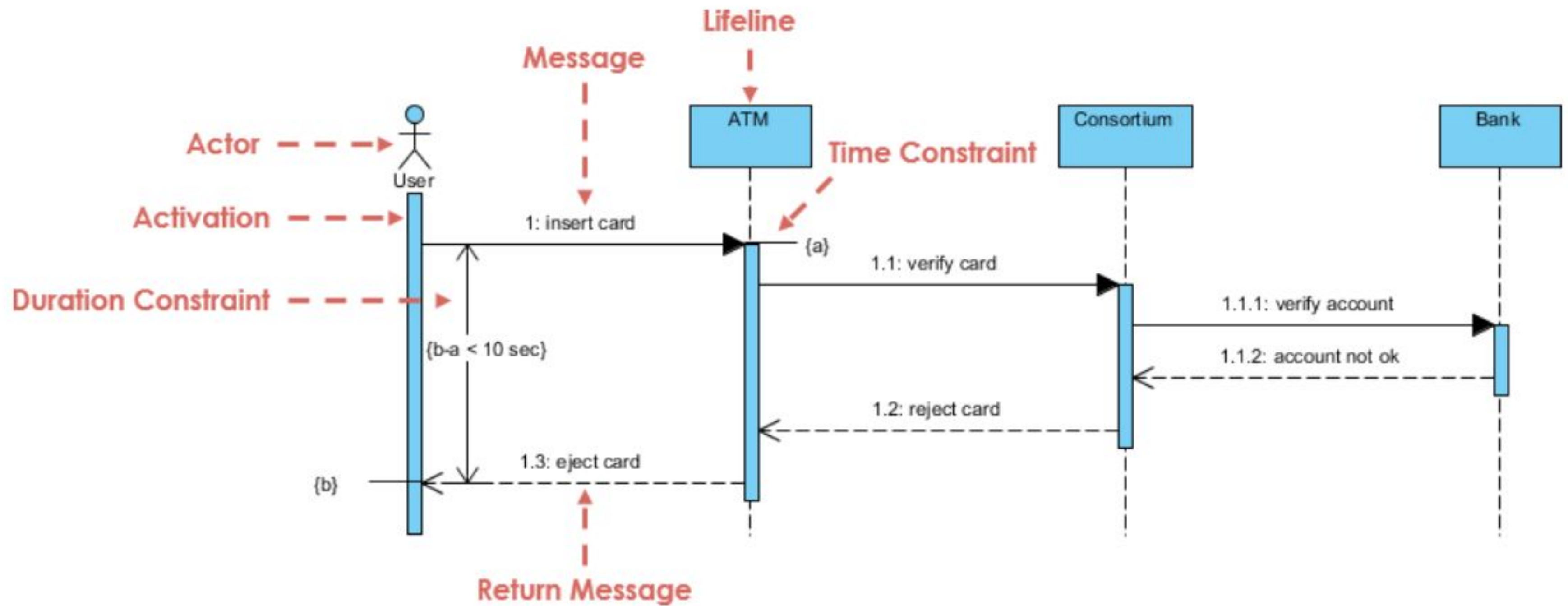


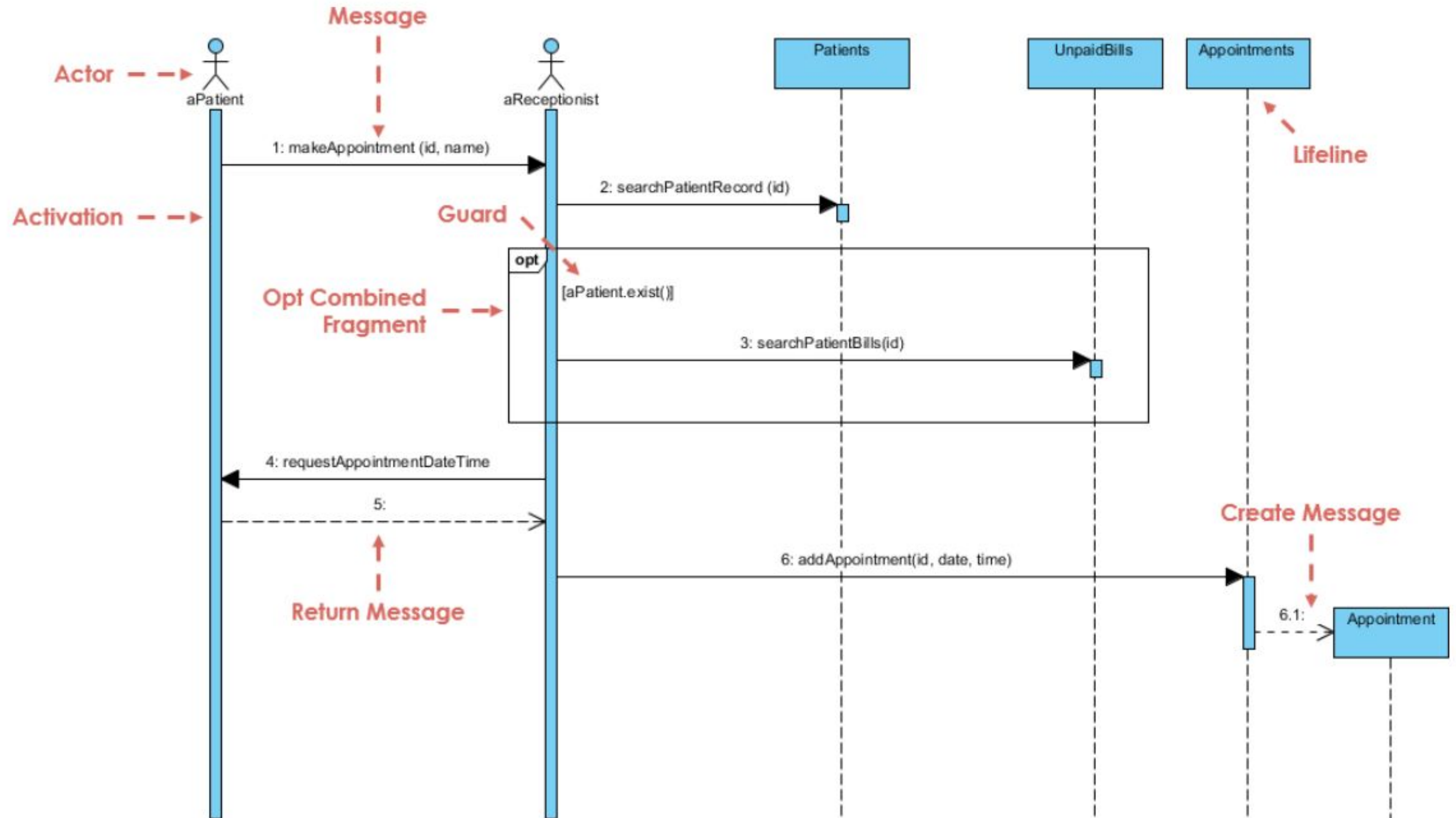


EXAMPLE









THE END